Ka-Kyung Kim, Ph.D.

Bioinformatics researcher linking biological data analysis with drug-discovery biomarkers

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Summary

Bioinformatician with about seventeen years of professional experience (excluding the doctoral period) in genomic, transcriptomic, single-cell, and clinical multi-omics data, working since 2004 across national research institutes, NGS companies, and a diagnostics company. Carries analysis through to the clinical reports and services that hospitals and clients use, and builds LLM and agent workflows that tie every claim to its evidence.

Highlights

Papers	20+ SCI(E) papers (first author, Sci Data 2019; co-author, Nat Commun 2020); **first and corresponding author, ML4H 2026 Findings (NeurIPS workshop)**
Talks	**Oral presentation accepted, GIW ISCB-Asia 2026** (22 of 287 abstracts)
Awards	**Winner, NVIDIA x FastCampus Korea Agentic AI Hackathon 2026** (team lead); Grand Prize, Korea Clinical Datathon 2019
Deployment	Cell image review assistant deployed in diagnostic work, **cutting the cells needing human confirmation by about 70%** (Cytogen)
Quality	**New GCLP accreditation** for a clinical specimen laboratory, with CAP proficiency testing twice a year and annual ISO27001 (Theragen Bio)

Focus Areas

- Multi-platform NGS and multi-omics analysis, including scRNA-seq
- Clinical trial data analysis (Ph1-3) and regulatory frameworks (GCLP/CAP/ISO)
- End-to-end pipeline development with automation and QC standardization
- AI/LLM-driven variant interpretation and analysis workflows (PyTorch, scikit-learn, LangChain)

Professional Experience

2025-present — **Cytogen** Chief Researcher, Diagnosis Division

Develops and deploys AI for diagnostic work, including a cell image review assistant, four in-house data tools on one internal portal, and shared criteria for AI tool use. Built the operating base for CTC diagnostics, the scRNA-seq pipeline, clinical report automation, and GCLP quality systems, and defined the diagnostic service portfolio and its pricing from market research.

2021-2024 — **Theragen Bio** Director · Clinical Analysis Team / Bioinformatics

Led Ph1-3 oncology trial analysis and served as analysis lead for the national bio-big-data pilot. Obtained new GCLP accreditation and designed its data management scheme, and oversaw an AWS-based pipeline for large-scale, fast-turnaround WGS delivery.

2018-2021 — **Macrogen** Deputy GM · Business Development / Clinical Analysis

Led K-MASTER genomic analysis, BRCA1/2 software with CE-IVD documentation, and liquid-biopsy commercialization strategy. In Business Development (2021), assessed the feasibility of licensing an overseas genomic analysis service and set analysis-service strategy by segment.

2015-2018 — **Yonsei University** Postdoctoral Researcher · Biomedical Systems Informatics

Discovered disease variants from multi-omics data and ran non-coding eQTL translational research on gastric cancer PDX, canine mammary tumors, pediatric allergy, and neurological disease.

2012-2015 **SAIHST, Sungkyunkwan University** Postdoctoral Researcher · Molecular Informatics and Genomics

Built the miRTCat database and, as an NRF Research Fellow, studied disease variants at non-canonical microRNA target sites.

2007-2012 **Samsung Biomedical Research Institute** Researcher · Human Genomics (2007-2011 concurrent with the Ph.D.)

Analyzed the Korean HapMap and KARE national cohorts, covering pharmacogenomics and genotype imputation for the Korean population.

2004-2007 **Korea National Institute of Health** Research Technician · Center for Genome Science

Discovered disease-associated variants in the Korean population and built and operated the SNP database.

Selected Programs

Cytogen **CTC-based liquid biopsy diagnostics**

Built and commercialized end-to-end operations for circulating-tumor-cell diagnostics: assay concept, pipeline, clinical report, and customer technical support.

Theragen Bio **Immuno-oncology clinical trial analysis, Ph1-3**

Led oncology trial analysis for three pharmaceutical companies.

Theragen Bio **Large-scale WGS delivery on AWS**

Oversaw an AWS-based cloud pipeline for high-volume, fast-turnaround WGS delivery to pharma clients.

Macrogen **K-MASTER precision oncology genomics**

Led genomic analysis for a national precision-oncology platform.

Macrogen **BRCA1/2 variant analysis software, CE-IVD documentation**

Directed development of software spanning variant calling, clinical interpretation, and automated report generation, together with CE-IVD-grade documentation and performance evaluation.

Theragen Bio **Quality and regulatory systems**

Obtained new GCLP accreditation for a clinical specimen analysis laboratory, and established recurring certification cycles: CAP proficiency testing twice a year and ISO27001 annually.

Built

In-house · Vision **Cell image review assistant**

A classification model for CTC image review trained on 1,531 labeled cells, with specimen-level cross-validation to block data leakage (an inflated kappa of 0.910 was corrected to 0.704 before release). Confidence-based triage cut the cells needing human confirmation by about 70%. Trained on GPU and deployed for CPU inference in-house.

In-house · Tooling **AI usage criteria, four data tools, and an internal portal**

Compared three AI tools under the same conditions and set criteria for which tasks each fits. Built four tools from internal requirements for review, laboratory, and sales staff: a test status board, a report generator, an analysis dashboard, and a literature search tool, all on one internal portal. The dashboard answers questions through retrieval (RAG) and shows the source documents with each answer.

In-house · LLM **LLM-based genomic Q&A and clinical report automation**

An LLM system that answers from test results and source documents (RAG), together with clinical report automation.

Personal · Validation

ADMET split audit

A benchmarking study of how the train/test split changes reported performance. Compared random, scaffold, and structure-disjoint splits on 12 TDC ADMET tasks and 3 pharmacokinetic tasks, using fingerprint+LightGBM, a message-passing neural network, and a fine-tuned ChemBERTa. Under the structure-disjoint split all 12 tasks lost performance (median drop 0.098, 5 seeds). With 15 seeds, the top-10% hit rate fell in all 8 tasks evaluated, from 0.358 to 0.176 for volume of distribution (VDss, $p<0.001$). Three initial conclusions were retracted after adding seeds, and the correction was published openly.

Personal · LLM

VeriVar

A somatic variant interpretation pipeline that controls unsupported citations with citation-grounded retrieval and a faithfulness contract. It scores ClinGen/CGC/VICC 2022 oncogenicity and AMP/ASCO/CAP 2017 clinical tiers together, and runs rules, an LLM, and a no-retrieval baseline side by side. Presented at ABDD Summit 2026 (i-Talk); repository release in preparation.

Multi-agent research

AutoBioX

A multi-agent research harness that splits data preparation, prediction, verification, evidence search, and manuscript drafting. Project lead, working with six researchers from other institutions through Pseudo Lab. Its per-gene multiome RNA velocity reliability map was accepted to the ML4H 2026 Findings track (NeurIPS 2026 workshop; first and corresponding author) and selected for an oral presentation at GIW ISCB-Asia 2026.

Hackathon · Multi-agent

FlyGate

A drug-candidate verification agent that catches claims going beyond the evidence they cite. It links binding prediction (DiffDock NIM), measured affinity, approved labels, adverse-event reports (FAERS), and literature in one flow, and attaches an evidence ID to every claim. Author and critic workflows run separately on NeMo Agent Toolkit; 15 over-interpretation rules plus model judgment rejected all 16 planted over-claims with 0 of 17 false positives, and an OpenShell deny-by-default policy blocks access outside allowed hosts. Team lead of five; owned the agent harness, pharmacovigilance tools, sandbox policy, and evaluation. Won the NVIDIA x FastCampus Korea Agentic AI Hackathon 2026, first among 5 finalist teams (from 10 at the on-site round) at NVIDIA AI Day Seoul on 2026-11-10. github.com/Team-FlyGate/Project-FlyGate

Personal · MCP

MCP servers and agents

PharmaSignal gathers regulatory filings and clinical trial registrations to organize drug safety and competitive signals, and routes inference by data sensitivity so restricted material never reaches a commercial API (Agent Forge AI Hackathon Seoul, 2026.08; github.com/kakyungkim/pharmasignal). ChillMCP is an MCP server on FastMCP with 12 model-callable tools, 5 interactive prompts, state handling, and a test client, released under MIT (SKT AI Summit Hackathon 2025 pre-task; github.com/kakyungkim/chillmcp).

AI agent · Hackathon

Vital STORM

An agent that grades key health-screening values on a five-level scale and recommends follow-up examinations, answering from four national health-screening manuals through retrieval (RAG) (Instruct.KR x Sonic 2025 hackathon).

Open source · Agent harness

econ-radar

An agent harness that assembles and publishes a daily economic digest weighted toward pharma, bio, and AI.

Selected Publications

2020

Cross-species oncogenic signatures of breast cancer in canine mammary tumors

Kim TM, … Kim KK, … Kim S. Nature Communications 11:3616

2019 Whole-exome and whole-transcriptome sequencing of canine mammary gland tumors First author
Kim KK, Seung BJ, ... Kim S. Scientific Data 6(1):147

2018 Palmitate and minimally-modified low-density lipoprotein cooperatively promote inflammatory responses in macrophages Co-first author
Ann SJ†, Kim KK†, ... Miller YI. PLoS One 13:e0193649

2013 miRTCat: a comprehensive map of human and mouse microRNA target sites including non-canonical nucleation bulges First author
Kim KK, Ham J, Chi SW. Bioinformatics 29(15):1898-1899

The full list of 20+ papers is on Google Scholar. Workshop: accepted to the ML4H 2026 Findings track (NeurIPS 2026 workshop), "A reliability map for per-gene multiome RNA velocity parameters in single-cell kinetics", first and corresponding author. Conference abstracts: two abstracts were accepted at GIW ISCB-Asia 2026, one oral and one poster; the poster is joint AutoBioX work, "Predictability Is Not Substitutability: A cost-of-substitution framework for H&E-based molecular prediction across 5 cancers".

Talks

2026.11 **A reliability map for per-gene multiome RNA velocity parameters in single-cell kinetics**
BIOINFO 2026 · GIW XXXV · ISCB-Asia, Seoul · oral presentation (accepted; 22 of 287 abstracts)

2026.11 **Multi-agent workflows for biomedical research: research agents grown at Pseudo Lab**
AWS Community Day 2026, Seoul · community talk (upcoming)

2026.07 **The AutoBioX multi-agent pipeline**
PseudoCon 2026 · talk

2026.01 **Automated Clinical Reporting for Oncology NGS: Dual-Mode LLM Pipeline**
ABDD Summit 2026 at Stanford · i-Talk and poster

2025.11 **Single-cell QC standardization and an scRNA-seq × immunofluorescence pipeline**
Future of Medicine Symposium: At the Crossroads of Computational Pathology and Spatial Biology · lightning talk

2024.10 **Poster presentation, ABDD Summit**
Awarded Best Poster Presentation

2022.04 **Cancer: a history of the emperor of all maladies**
1st K-BioX Global Mentoring EXPO · invited talk

Writing

2026 **Generative AI in biotech and drug discovery: technology, cases, and ecosystem outlook** BRIC View 2026-T24
A trend report separating how strong the evidence is at each stage, from protein design through the clinic.

2026 **Integration and expansion of bio big data: data, AI, and industry** BRIC View 2026-T11

2025 **Single-cell analysis in drug discovery and patient selection** BRIC View 2025-T29

2018 **Discovery trends and outlook for allergic disease therapeutics** BRIC View 2018-T16

2017 **Cloud-based management and services for genomic information** BRIC View 2017-T14

Awards and Competitions

2026	NVIDIA x FastCampus Korea Agentic AI Hackathon 2026, Winner · team lead, Team FlyGate; first of 5 finalist teams at NVIDIA AI Day Seoul (2026-11-10); prize: NVIDIA DGX Spark
2026	KNIH-KSBi Joint Healthcare AI Challenge 2026, finalist · passed the written review; final presentation on 2026-11-17 (Korea National Institute of Health and KSBi)
2025	Cheiron ResearchThon, Creative Academic Impact Award · liquid-biopsy multi-omics model for tumor microenvironment prediction
2024	ABDD Summit, Best Poster Presentation Award · Merit Awards
2021	Medical Informatics Advanced Course, Encouragement Award · Asan Medical Center
2019	Korea Clinical Datathon, Grand Prize · sepsis prediction, KoNECT
2013	KSBMB Annual Meeting, Excellent Researcher Award (poster)

Service and Community

2026	Reviewer, ML4H 2026 (Machine Learning for Health, NeurIPS 2026 workshop)
2026-	Project lead, Falsify (Pseudo Lab, Season 13): a benchmark that plants defects in open-access biomedical papers and builds the tools that catch them

Education

2021-2023	**Korea National Open University** B.S. Health & Environment, Business (double major)
2007-2011	**Sungkyunkwan University** Ph.D. Molecular Medicine
2002-2004	**Korea University** M.S. Life Science
1996-2000	**Dongguk University** B.S. Computer Engineering

Certifications and Languages

- NVIDIA Certified Associate: AI Model & Application Development (NVIDIA, 2026); Engineer Information Processing (HRD Korea, 2006); Linux Master Level 2 (KAIT, 2006)
- English: OPIc IM2; reads and writes scientific papers and technical documents; one year of language study in the United States
- Machine learning and data analysis: Coursera (Johns Hopkins Univ.), KDT AIFFEL Data Scientist course (2024)
- AI-driven drug discovery data analysis and SaMD model development (KOHI and KSA, 2025); Biohealth AX Executive Strategy Program (KOHI, Aug-Sep 2026)

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